

Affan Bin Usman

Santa Clara, CA | (602) 756-3989 | affanbu25@gmail.com | [LinkedIn](#) | [GitHub](#) | [Website](#) | [Google Scholar](#)

Summary

Machine learning research engineer with 2.5+ YOE designing, building, and deploying end-to-end ML systems. Proficient in the full ML lifecycle - research, data pipelining, feature engineering, model development, deployment & optimization. Passionate about deep learning and perception for robotic applications, to deliver robust and scalable AI solutions.

Technical Skills

Languages & Tools: Python, C++ (familiarity), Bash, Git, Docker, Kubernetes, Google Cloud, Azure, BigQuery/SQL

ML Frameworks: PyTorch, TensorFlow, PyTorch Lightning, Hugging Face, LangChain

MLOps & Deployment: CI/CD, Real-Time Inference, ETL Pipelines (Airflow), Data Caching (HDF5), ROS2

AI/ML: Deep Learning (CNN, LSTM), Generative AI (Stable Diffusion, GANs), LLMs (Prompt Engineering, Fine-Tuning), Computer Vision (OpenCV, YOLO), NLP, Reinforcement Learning, Transformers

Professional Experience

Gatik AI | Mountain View, CA

Machine Learning Engineer

July 2025 – Present

- Bridging research and production models for autonomous vehicles by prototyping advanced perception and semantic models to improve existing architecture and guide strategic planning for continued improvement.
- Simplified model performance comparisons by engineering a scalable evaluation framework with in-memory representation for perception (classification & detection), tracking & latency metrics, enabling granular analyses.
- Implemented and benchmarked SOTA open-source perception models to evaluate their feasibility and performance against internal baselines.

Honda Research Institute USA | San Jose, CA

Research Engineer (Machine Learning)

Dec 2023 – Mar 2025

- Engineered and deployed a real-time (15 FPS) multimodal perception system for a social robot (Human-Computer Interaction) by integrating a CNN/LSTM model with ROS2, Intel RealSense cameras, and microphones.
- Optimized a complex ML training pipeline by 30% by implementing distributed data processing and model parallelism with PyTorch Lightning and HDF5 data caching strategies.
- Enhanced model recall by 21% on a highly imbalanced dataset through advanced audio-visual feature engineering, dimensional reduction, normalization strategies, weighted sampling, and cross-entropy loss function.
- Developed and fine-tuned an LLM-powered agent using OpenAI's API to enhance behavioral understanding, directly improving robot's interactive capabilities.
- Co-led design and execution of a large-scale user study (105+ participants) in collaboration with Cornell, to build a novel multimodal dataset, enabling key advancements in human-robot interaction.

Jenius Bank | Scottsdale, AZ

Data Engineer (Machine Learning)

Sep 2023 – Dec 2023

- Automated end-to-end ETL workflows using Apache Airflow and Google BigQuery, reducing manual data integration effort by 20% and providing clean data for downstream analytics and ML models.

Interactive Robotics Lab, ASU | Tempe, AZ

Machine Learning Research Intern

May 2023 – Sep 2023

- Developed and optimized an LSTM-based gesture recognition model, accomplishing 85% classification accuracy on a custom PPG sensor dataset through rigorous hyperparameter tuning.

Publications & Patents

- Usman, A., et al. (2024). "Modeling interpersonal perception in dyadic interactions: towards robot-assisted social mediation." *Frontiers in Robotics and AI*.
- Co-inventor on US Patent Application (2024): "Graph-Based Modeling Of Relational Affect in Group Interactions".

Education

Arizona State University (ASU) | Tempe, AZ

May 2023

Master of Science in Robotics & Autonomous Systems (Artificial Intelligence Concentration) | GPA: 4.0/4.0

National University of Sciences & Technology (NUST) | Islamabad, Pakistan

May 2019

Bachelor of Engineering in Electrical Engineering

Academic Projects

Image Generation | Generative AI, Stable Diffusion, PyTorch, CNNs

- Fine-tuned a Stable Diffusion model on a custom dataset of 100+ images, engineering a pipeline including data augmentation and model training; achieved a 91% conformity rate for generated images.

Predictive Path Planning for Dynamic Environments | GANs, PyTorch, Robotics, Motion Planning (RRT*)

- Integrated a Social-GAN with an RRT* path-planning algorithm to create a predictive navigation system for a mobile robot; succeeding with a 70% success rate in avoiding collisions with six dynamic obstacles.

Autonomous Vehicle Perception System | Computer Vision, YOLOv4, OpenCV, Python

- Built and trained a YOLOv4 model on a custom dataset of 550+ annotated images for real-time object detection; attained 90% detection accuracy with model tested on a miniature vehicle.

Image Captioning with Attention | NLP, ResNet, LSTM, Attention Mechanisms, PyTorch

- Revamped performance by 80% on SacreBLEU score compared to a baseline model by applying transfer learning on an encoder-decoder model with an attention mechanism for image captioning with Flickr8K and GCC datasets.

Reinforcement Learning for Robotic Path Planning | Reinforcement Learning, Q-Learning, Python

- Implemented a Q-learning algorithm from scratch to train an agent for optimal navigation in a grid world, improving path planning efficiency by 45% by modeling a custom reward function and epsilon decay.

Multi-Robot Exploration & Mapping | Robotics, Simulation, Path Planning, Python

- Designed a multi-robot simulation for search-and-rescue, attaining 100% mapping accuracy using potential gradient obstacle avoidance and coordinated motion strategies (Levy walk and Brownian motion).

Leadership & Additional Experience

Arizona State University | Tempe, AZ

Faculty Associate

August 2023 – December 2023

- Taught engineering fundamentals to 150+ students (focus: engineering design, research, prototyping, and coding)

Pakistan Oxygen Limited | Pakistan

Assistant Plant Manager & Process Engineer (Management Trainee)

March 2020 - November 2021

- Drove plant efficiency by optimizing resource utilization and standardizing maintenance procedures, resulting in an 11% increase in CO₂ production capacity and 100% power uptime.

Sales Engineer (Management Trainee)

September 2019 - February 2020

- Exceeded monthly sales target by 200% by leveraging market segmentation and opportunity analysis to identify and secure new business.